

Appendices

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Appendix A - Derivation of Relative Growth Values from Official National Accounts Data

*Some countries analyzed in the ISIC dataset lack data for all of the years listed in our general analysis of the dataset (2005-2022). The countries without full data availability are listed below, those not mentioned had data for all years: Antigua and Barbuda: 2012-2022, Bahamas: 2013-2022, Barbados: 2011-2022, Cabo Verde: 2008-2022, Cook Islands: 2007-2022, Fiji: 2012-2022, Guyana: 2007-2022, Mauritius: 2007-2022, Papua New Guinea: 2007-2018, Saint Lucia: 2007-2020, Samoa: 2008-2020, Seychelles: 2015-2022, Suriname: 2016-2022, Trinidad and Tobago: 2013-2022,

In this Appendix, we describe the method(s) used to calculate relative growth values for economic sectors in a set of small island developing nations (SIDS). This model draws on the existing Growth Models literature, with previous work tabulating sectoral growth through the use of OECD input-output tables (Baccaro and Hadziabdic 2022). Due to the fact that input-output tables are not available for the SIDS, we have adapted this model to the readily available International Standard Industrial Classification of All Economic Activities (ISIC) data provided by the United Nations Statistics Division (Citation ISIC3, Citation ISIC4). The two revisions have slightly different series with varying combinations of sectoral components, making a merge of the two impractical. ISIC 3 is only used for the country of Jamaica, one of the case studies examined in the chapter. Jamaica has yet to update to the ISIC 4 revision and as such it was necessary to use the ISIC 3 data that they supply which spans from 1998-2023.

Both revisions are provided in local currency at the current price level in the form of excel spreadsheet with with columns for country, series, fiscal year and value (Table 1).

Table 1: Original ISIC Tables

Country or Area	Item	SNA93 Item Code	Fiscal Year	Value
Antigua and Barbuda	Agriculture, hunting, forestry; fishing	A+B	2022	44800000
Antigua and Barbuda	Fishing	B	2022	42600000
Antigua and Barbuda	Mining and quarrying	C	2022	27300000
Antigua and Barbuda	Manufacturing	D	2022	107800000
Antigua and Barbuda	Electricity, gas and water supply	E	2022	158000000
Antigua and Barbuda	Construction	F	2022	576200000

We first pivot the data into a “wide” format, where each item in the dataset is given its own column and

then computed variables from the original series (16 for revision 3 and 20 for revision 4) as follows (System of National Accounts codes listed in parentheses):

Revision 4:

Primary = Agriculture, forestry and fishing (A)
Industry = Manufacturing, mining and quarrying and other industrial activities (B+C+D+E)
Construction = Construction (F)
Transportation = Transportation and Storage (H)
Hospitality = Accommodation and food service activities (I)
Financial = Financial and insurance activities (K)
Real Estate = Real estate activities (L)
Other: Public = Wholesale and retail trade, Information and communication,
Professional, scientific, technical, administrative and support service activities,
and other service activities (G+J+M+N+R+S+T)
Other: Private = Public administration and defence, education,
human health and social work activities (O+P+Q)

Revision 3:

Primary = Agriculture, hunting, forestry; fishing, Mining and quarrying (A+B+C)
Manufacturing = Manufacturing (D)
Electricity, gas, water supply = Electricity, gas and water supply (E)
Construction = Construction (F)
Hospitality = Hotels and restaurants (H)
Transportation = Transportation (60-63)
Communications = Post and telecommunications (64)
Financial Intermediation = Financial intermediation (J)
Real Estate = Real estate, renting and business activities (K)
Other: Private = Wholesale, retail trade, repair of motor vehicles, motorcycles and personal and household goods
and Private households with employed persons (G+P)
Other: Public = Public administration and defence; compulsory social security,
education; health and social work; other community, social and personal services (L+M+N+O)

From these transformations we obtain a simplified table with columns more interpretable for analysis while retaining the level of granularity that led us to use these ISIC data in the first place (Table 2).

Table 2: Reduced Structure of ISIC Tables

Country or Area	Year	Primary	Industry	Construction	Transportation	Hospitality	Financial	Real Estate	Other: Private	Other: Public
Antigua and Barbuda	2012	0.09	-0.27	0.29	0.02	0.02	0.13	-0.05	0.35	0.41
Antigua and Barbuda	2013	-0.37	0.34	0.47	0.09	0.50	-1.72	-0.02	0.50	1.21
Antigua and Barbuda	2014	0.00	0.12	0.19	0.08	0.17	0.00	0.05	0.19	0.20
Antigua and Barbuda	2015	0.02	0.26	0.08	0.02	0.23	0.10	0.03	0.23	0.02
Antigua and Barbuda	2016	0.02	0.28	0.50	0.16	0.16	-0.38	-0.10	0.48	-0.11
Antigua and Barbuda	2017	0.09	-0.09	0.64	-0.16	-0.08	0.19	0.15	0.10	0.16

We then perform the following operations on these transformed datasets for every country year:

$$\text{Change in Total Value Added: } \Delta \text{TVA} = \text{TVA}_t - \text{TVA}_{t-1}$$

$$\text{Change in Sector Value Added: } \Delta \text{Sector} = \text{Sector}_t - \text{Sector}_{t-1}$$

$$\text{Absolute Growth Contribution: } \frac{\Delta \text{Sector}_t}{\Delta \text{TVA}_t}$$

$$\text{Relative Growth Contribution: } \frac{\Delta \text{Sector}_t}{\text{TVA}_{t-1}}$$